Application of Machine Learning in Dynamic Multi-Factor Stock Selection

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Abstract. The stock market has long been a crucial component of modern economic systems, significantly impacting the lives of both individual and institutional investors. Investment strategies have increasingly evolved from static to dynamic and adaptive approaches, thanks to advancements in data science, to navigate the inherent complexities of the market. This paper presents a dynamic multi-factor stock selection strategy enhanced by machine learning techniques, including clustering and dynamic factor weighting. By regularly updating the factors and their corresponding weights, this strategy aligns with the current market conditions, such as varying levels of volatility, market trends, and overall uncertainty. Additionally, the study includes comprehensive backtesting that demonstrates the superior performance of the machine learning-based dynamic strategy over a static multi-factor approach and the traditional S&P 500 index benchmark. Notably, the dynamic strategy achieved an annualized return (CAGR) of 47.57%, significantly exceeding the S&P 500's 14.41% and the passive strategy's 14.41%, while also delivering a more favorable Sharpe ratio, indicating improved risk-adjusted returns despite experiencing greater volatility and drawdowns.

Keywords: Machine Learning, Dynamic Multi-Factor Strategy, Stock Selection, Clustering Analysis, Portfolio Optimization.

1 Introduction

The stock market has been a significant part of the world's economic systems for many years and has affected investors and institutions. While conventional approaches to security selection, which are based on fundamental and technical analysis, may have specific limitations in that they may not capture the market dynamics, machine learning (ML) techniques can solve these limitations and design data-driven, dynamic, and adaptive investment policies [1, 2].

This paper proposes a Dynamic Multi-Factor Stock Selection Strategy based on machine learning algorithms. It uses clustering techniques and dynamic factor weighting to improve stock selection. This way, the factors and their corresponding weights are adjusted in real-time to better deal with market conditions such as volatility, bullish and bearish phases, and economic uncertainty than traditional static multi-factor models [3].

© The Author(s) 2025
Y. Sun (ed.), *Proceedings of the 2025 3rd International Conference on Image, Algorithms, and Artificial Intelligence (ICIAAI 2025)*, Advances in Computer Science Research 122,
https://doi.org/10.2991/978-94-6463-823-3_65

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The dynamic weighting of the factors across the different market phases increases the risk-adjusted returns by aligning with the market characteristics. This paper also applies backtesting to compare the dynamic strategy with traditional models, such as the S&P 500 index, to determine the effectiveness of this approach in real-world investment settings.

The strategy's first step is to gather financial data factors that include market returns, size, value, profitability, and investment style, as well as feature engineering to clean the dataset. The strategy also uses K-Means clustering and Gaussian Mixture Models (GMM) to identify market periods, such as low-risk, bull, bear, or volatile markets. These techniques show when factors and stocks are at their peak to help improve the stock selection process.

The primary purpose of this research is to develop a durable and dynamic stock selection model that can outperform static strategies in portfolio optimization and risk management. This approach provides a more adaptive and proactive way of dynamically addressing stock selection by using clustering to alter factors and market conditions.

2 Literature Review

Applying machine learning algorithms in stock selection models has greatly changed quantitative investment strategies. Machine learning provides better techniques for shaping extensive data and identifying the complex, non-linear relationships between financial factors. Machine learning approaches improve multi-factor models, which use various financial indicators to select stocks, in several essential ways.

2.1 Dynamic Multi-Factor Stock Selection Models

Li, Zhang, and Wang suggested a dynamic multi-factor stock selection strategy based on XGBoost, a gradient boosting algorithm, to predict the Information Coients (ICs) of different factors [4]. They also pointed out that the conventional factor weighting is static, which means that the importance of the factors is fixed over time and does not change according to the market dynamics. They arrived at the conclusion that XGBoost is better than the traditional fixed-weight models, and there is a significant improvement in the stock selection efficiency.

Similarly, Gao et al. applied a recurrent neural network approach in order to improve the forecast of stock returns and thus improve the temporal analysis of stock prices [5]. Their model captures the time-varying correlations between stocks, which is crucial for real-world stock selection strategies where market dynamics change over time. They were, therefore, able to use advanced deep learning methods to solve the problem of non-stationarity in stock markets.

In a comprehensive study, Wei, Dai, and Lin, in a wide-ranging study, suggested an end-to-end deep learning framework for active investing (E2EAI), which encompasses all the phases from factor identification to portfolio formation [6]. Their framework uses gated attention mechanisms for dynamic factor selection and stock weighting and thus provides a unified and structured approach to factor-based stock selection. The

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E2EAI model guarantees that all the aspects of the stock selection process are aligned to the single investment goal, thus optimizing the portfolio returns and risks.

2.2 Machine Learning Algorithms for Stock Selection

The effectiveness of machine learning algorithms such as Random Forests, Support Vector Machines, and XGBoost in stock selection is well-documented. Zhong et al. compared multiple machine learning algorithms, including XGBoost, in a multi-factor stock selection framework, demonstrating that XGBoost provided superior performance in predicting stock returns over other models [7]. XGBoost's ability to handle large datasets and its robustness in predicting stock returns across different market conditions make it an ideal choice for dynamic factor-based stock selection.

Additionally, Fu et al. proposed a sustainable stock selection strategy that utilized the Random Forest (RF) algorithm for dynamic factor adjustment, highlighting its ability to predict stock returns more reliably than traditional methods [8]. Their study also underscored the importance of continuously adjusting the factors used in stock selection to reflect changing market conditions, an approach that aligns with the evolving nature of financial markets.

2.3 Summary

However, there are challenges in adopting machine learning-based stock selection models. A central issue is the uncertainty of financial factors that are constantly changing. This paper has contributed to developing a dynamic multi-factor approach that uses clustering techniques such as K-Means and GMM to identify market regimes and adjust the strategy. This method enables more flexible stock selection. Furthermore, to enhance the understandability of the model, this paper focuses on factor exposure visualization and clustering analysis to better understand the decision-making process for improved transparency.

3 Feature Engineering

Feature engineering is a critical step in developing a robust dynamic multi-factor stock selection strategy. It involves several key steps, including data acquisition, preprocessing, calculating factor exposure, visualizing the factor exposure, and performing clustering analysis [9].

3.1 Data Acquisition

The purpose of data collection in this study is to collect the necessary data on stock returns and market factors. The first step is to download the monthly closing prices of the components of the S&P 500 using the 'yfinance' library. The date range of the stock data matches the Fama-French factor data, available from July 2012 to January 2025. It is crucial to avoid using stock and factor data collected at different times, as this may

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result in incorrect factor exposure calculations. Furthermore, the dataset has multi-index data, but only the "Close" prices of the stocks are used in the analysis. This ensures that the data remains consistent and can be effectively utilized for factor exposure analysis.

Furthermore, the Fama-French five-factor data is downloaded, encompassing factors such as Excess Returns (Mkt-RF), Size (SMB), Value (HML), Profitability (RMW), and Investment (CMA) using the 'requests' library. Then, factor data is downloaded and processed in the desired format to be consistent with the stock data. The time indices of the stock and factor data are matched to enable accurate factor exposure determination and portfolio optimization [10, 11].

3.2 Data Preprocessing

Data preprocessing is essential to ensure the uniformity and accuracy of the datase3t, allowing for effective factor exposure calculations. The first task is to compute the monthly stock returns. The return for each stock at time t is calculated using the following formula.

$$Return_t = \frac{Price_t - Price_{t-1}}{Price_{t-1}} \tag{1}$$

Where $Price_t$ is the stock's closing price at time t, and $Price_{t-1}$ is the price at the previous time point $t-1$. This formula computes the percentage change in the stock's price from the previous month. Any stocks with missing data points are removed to ensure the dataset's completeness.

Once the monthly stock returns are computed, the next step is to align the time indices between the stock returns and the factor data. This alignment ensures that both datasets have consistent time periods, which is crucial for the regression analysis in the subsequent steps. Any time points where the data does not match between the two datasets are discarded.

Additionally, the study uses forward and backward means imputation to handle any missing values in the factor exposure data. Missing factor exposure values are replaced by the average of the previous and next available values, as described by the following formula.

$$X_{new} = \frac{X_{previous} - X_{next}}{2} \tag{2}$$

Where $X_{previous}$ and X_{next} represent the previous and next available factor exposure values, respectively, and X_{new} is the imputed value. This method ensures that missing data points are reasonably filled, maintaining the integrity of the dataset for factor exposure calculations [12]. These preprocessing steps are crucial for ensuring that the data used in the factor exposure analysis is both consistent and complete, providing a solid foundation for the subsequent modeling and clustering steps.

3.3 Factor Exposure Calculation

Factor exposure is a critical component in this study, as it quantifies the sensitivity of stock returns to various macroeconomic and financial factors [11, 13]. The calculation

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of factor exposures involves a series of steps, which are outlined below. These exposures provide valuable insights into how each stock responds to changes in different risk factors, such as market risk, value, size, profitability, and investment strategies. The main steps of Factor Exposure Calculation will be elaborated in the following paragraphs.

The first step in calculating factor exposures is to compute the excess returns for each stock. Excess return is defined as the difference between the stock return and the risk-free rate. This allows the study to isolate the portion of returns that is attributable to the stock's performance, excluding any effects due to the risk-free asset. The formula for excess return is expressed as follows.

$$R_{i,t} - R_f = StockReturn - RiskFreeRate \tag{3}$$

Where $R_{i,t}$ refers to the return of stock i at time t, and R_f is the risk-free rate for the corresponding period.

The second step is to match each stock with the relevant data factor. The factors used in this study include the Market excess return ($Mkt - RF$), Size (SMB), Value (HML), Profitability (RMW), and Investment (CMA). These factors are commonly used in financial modeling to explain the variation in stock returns. Each of these factors represents a different aspect of the stock's exposure to market conditions and macroeconomic trends. The data is obtained for the same period to ensure consistency in the analysis.

The third step involves performing multiple regression analysis using the equation

$$R_{i,t} - R_f = \beta_0 + \beta_1 \cdot (Mkt - RF)_t + \beta_2 \cdot SMB_t + \beta_3 \cdot HML_t + \beta_4 \cdot RMW_t + \beta_5 \cdot CMA_t + \epsilon_t \tag{4}$$

In this equation, $R_{i,t}$ represents the return of stock i at time$t$, and R_f is the risk-free rate. $(Mkt - RF)_t$ signifies the market risk premium, while SMB_t, HML_t, RMW_t, and CMA_t are the size, value, profitability, and investment factors, respectively. β_0 is the intercept, indicating the abnormal return.$\beta_1, \beta_2, \beta_3, \beta_4$, and β_5 are the regression coefficients, representing the factor exposures, and ϵ_t is the error term.

Finally, the regression coefficients $\beta_1, \beta_2, \beta_3, \beta_4, \beta_5$ are extracted from the regression model. These coefficients represent the exposure for each stock, indicating how much the stock's return is influenced by each of the macroeconomic factors. A higher β value indicates that the stock has a stronger exposure to the corresponding factor. These coefficients are crucial for understanding the risks and opportunities associated with each stock, and they are used in further stock selection and portfolio optimization processes.

This method facilitates the identification of stocks that exhibit strong exposures to favorable factors, which are pivotal in constructing a well-optimized portfolio [10, 11].

3.4 Clustering Analysis

Clustering analysis is employed to effectively categorize stocks based on their factor exposures. This analysis was performed after having standardized the factor exposures using the Standard-Scaler method to ensure that all the input data were on the same scale and that outliers or factors with very large or small scaling would not unduly influence the analysis. After the data has been made to have zero meaning and unit variance through standardization, the K-Means and Gaussian Mixture Model (GMM) are used to discover natural groupings in the data. These clustering methods work by

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grouping stocks according to some similar measure, which allows for the recognition of different market regimes. The concept of K is important in defining the market classification with a high level of granularity. For different values of K, silhouette scores are calculated to find the best number of clusters. The silhouette score is a quantitative measure of the quality of a clustering, which tells how well a stock lies within its group compared to the other groups. The performance of both K-Means and GMM clustering methods is investigated with various cluster sizes to determine the most appropriate partition of the market according to factor exposure [14, 15].

3.5 Clustering Results

The clustering analysis results using both K-Means and Gaussian Mixture Models (GMM) are shown in Figure 1, Figure 2, Figure 3, and Figure 4, which show the clustering results for various values of K. The stock data is clustered using two different values of K for the K-Means method, namely K = 3 and K = 4. The K = 3 clustering outcome groups the stocks into three categories based on their factor exposures, and this is depicted in the scatter plots where different colors represent different stock categories. This configuration captures the main market regimes and the different patterns of stock behavior concerning factors. When K=4, the clustering is more fine-grained, and four categories are obtained. However, this segmentation may result in overfitting, i.e., the market regime classification is over-complicated for no significant increase in the clustering quality.

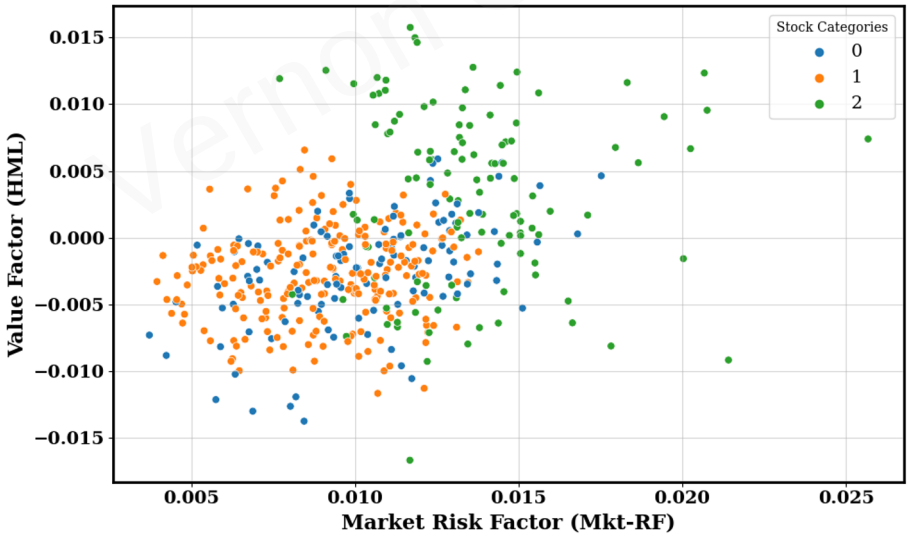

Fig. 1. GMM Clustering (K=3) - Factor Exposure

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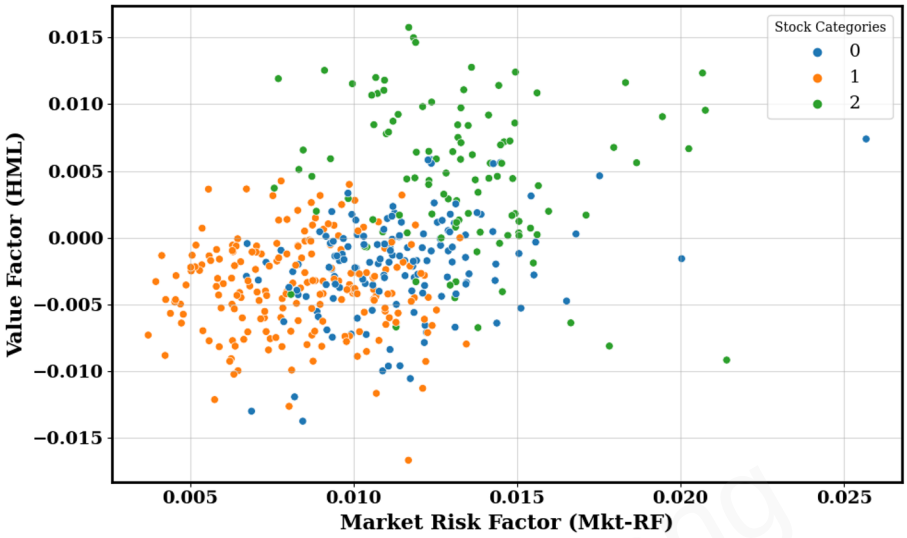

Fig. 2. K-Means Clustering (K=3) - Factor Exposure

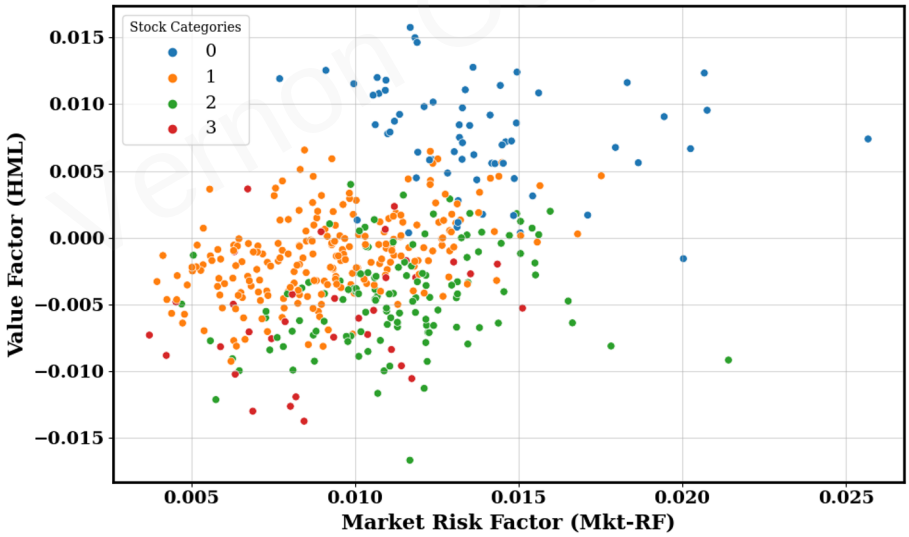

Fig. 3. GMM Clustering (K=4) - Factor Exposure

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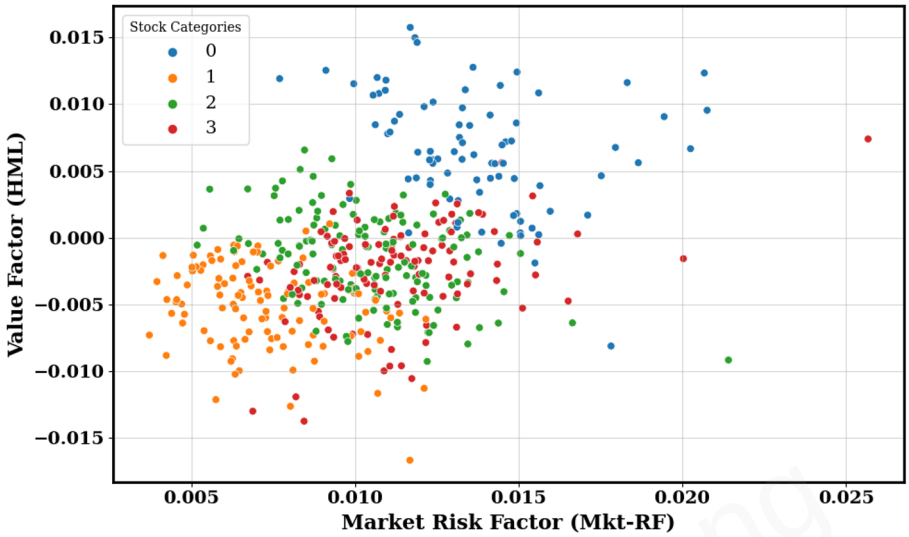

Fig. 4. K-Means Clustering (K=4) - Factor Exposure

The GMM clustering method, which allows for soft clustering, yields similar results but different interpretations, particularly when K=4, where the soft cluster boundaries offer a more graded perspective on market action. To determine the clustering quality, the silhouette scores are computed for K-Means and GMM. These scores state that K=3 appears to be the optimum clustering performance, as shown in Table 1 below, which compares the silhouette coefficients for both clustering methods for various values of K [15].

Table 1. Silhouette Coefficients for K-Means and GMM

Clustering Method	K=3	K=4
K-Means	0.2224	0.2048
GMM	0.1591	0.1386

These results further support K=3 as the ideal number of clusters because it offers a good middle ground between identifying distinct market regimes and over segmenting the market. The clustering results are useful in understanding the market status and steering the subsequent portfolio optimization and dynamic factor weighting strategy.

3.6 Key Findings from Clustering

In this study, clustering analysis was performed to determine several important market conditions that should be used for stock selection. When K=3, the analysis revealed three primary market conditions: Low-Risk Market, Bull Market, and Bear/Volatile Market. These conditions capture the primary states of the financial markets and provide a clear distinction between peaceful and turbulent times. This segmentation is in

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line with financial theory, which states that markets typically move between these major phases in response to changes in economic factors. The K=3 clustering solution was considered optimal because it provided a good tradeoff between the level of detail and simplicity relevant to the actual financial markets. When K=4 was used for clustering, there was over-segmentation, leading to low stability and transparent market transitions. The four clusters did not represent distinct and economically meaningful financial market phases that could be useful for informing dynamic investment policies. Furthermore, when K-Means and Gaussian Mixture Models (GMM) were compared, it was seen that GMM provided a better representation of the transitions between market regimes because it is a soft clustering method and is more appropriate for modeling the dynamics of financial markets [16]. In Conclusion, the K=3 configuration was considered the best for determining reasonable market regimes that are easy to understand and can be used in practice when optimizing a portfolio.

3.7 Silhouette Coefficients Evaluation

The silhouette coefficient is crucial in assessing clustering quality and determining the optimal number of K-Means and Gaussian Mixture Models (GMM) clusters. This coefficient evaluates how well a data point fits within its assigned cluster compared to other clusters, with higher values indicating better clustering. This paper computes the silhouette scores for various values of K to analyze the clustering outcomes. Results from both clustering algorithms indicate that K=3 yields the best clustering performance. As illustrated in Figure 5 and Figure 6, the silhouette scores for both K-Means and GMM peak at K=3, suggesting that this is the most effective value for characterizing the data structure. Beyond K=3, the silhouette scores begin to decline, indicating a deterioration in clustering quality as the number of clusters increases. This reduction in silhouette scores further supports the selection of K=3 as the optimal number of clusters, effectively distinguishing different market regimes without unnecessarily complicating the model. The silhouette score SS is defined as follow.

$$S = \frac{b - a}{max(a, b)} \tag{5}$$

Where a is the average distance from a point to other points within the same cluster, and b is the average distance from the point to the points in the nearest cluster. Notice that the silhouette coefficient SS ranges from -1 to 1, with values closer to 1 indicating better clustering performance.

These results are in line with prior studies, which also highlight the effectiveness of K=3 for financial market clustering. Tang and Zhang argue that using three clusters to categorize market regimes—such as low-risk, bull, and bear markets—provides a clear and intuitive segmentation of financial markets, aligning with the typical dynamics observed in historical market data [17]. Similarly, Lu found that applying K=3 in clustering methods for trading strategies in the U.S. stock market helps to identify distinct market phases, improving the accuracy of market regime predictions and the performance of trading models [18]. Therefore, the silhouette evaluation not only validated the clustering results but also reinforced the choice of K=3 as the most appropriate for this financial modeling approach.

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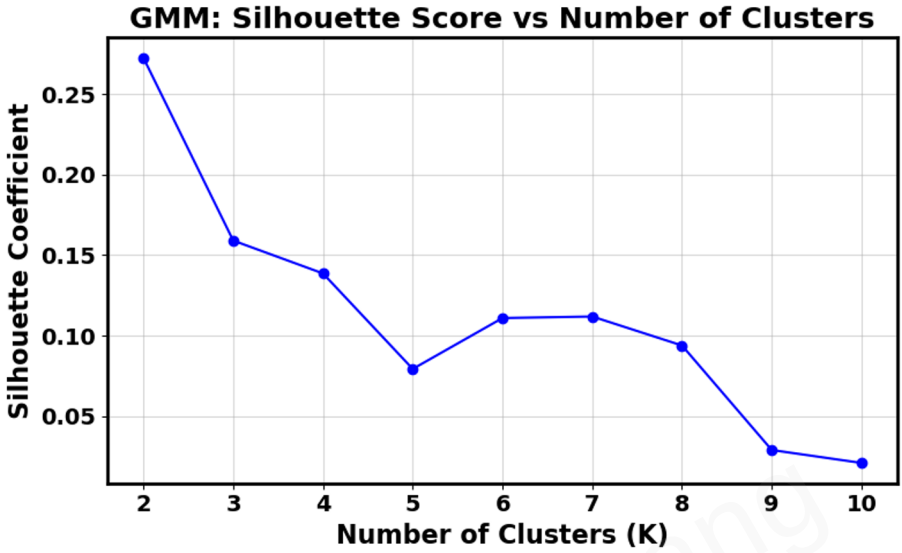

Fig. 5. Silhouette scores for GMM across different values of K

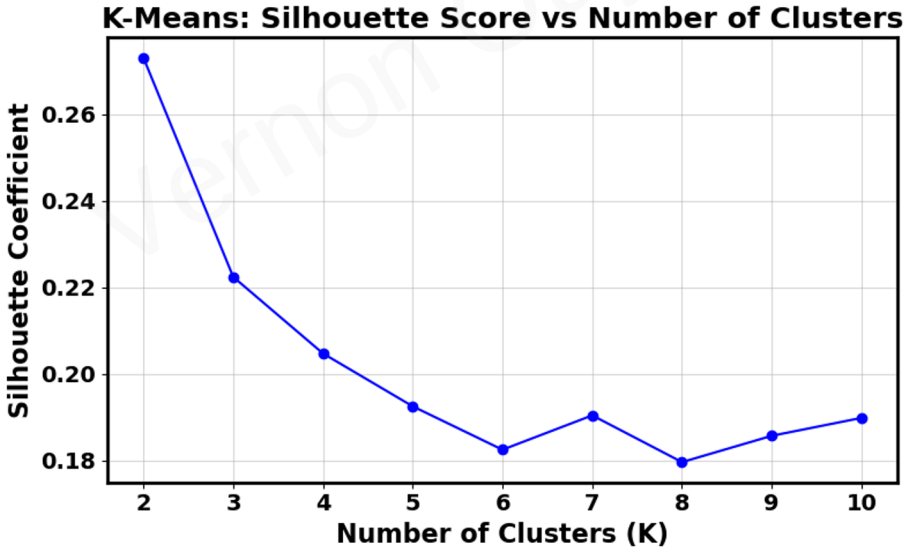

Fig. 6. Silhouette scores for K-Means across different values of K

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4 Market Period Clustering

4.1 Data Preprocessing

The preprocessing step is fundamental to ensuring accurate clustering results, as it involves extracting relevant market factors and standardizing the data. These key factors include the market risk premium (Mkt-RF), small-minus-big (SMB), high-minus-low (HML), robust-minus-weak (RMW), and conservative-minus-aggressive (CMA) factors. To standardize these variables, each factor was scaled to ensure that all inputs are on the same scale before being fed into the clustering algorithms. This transformation ensures that the analysis is not biased by the differing magnitudes or units of the variables. Standardization also allows for more consistent clustering results across different market periods.

4.2 Choosing the Best Clustering Method

To classify the market periods, this study employed both K-Means and Gaussian Mixture Model (GMM) clustering techniques. A crucial step in selecting the optimal clustering approach was determining the ideal number of clusters, denoted by K. To achieve this, the silhouette score was computed for various values of K. Silhouette scores are a widely recognized metric for evaluating the consistency of data points within clusters compared to other clusters, with higher scores indicating better-defined clusters [19].

The results indicated that K=3 was the most appropriate number of clusters for this study. This conclusion aligns with the inherent structure of financial markets, where three distinct market phases—low-risk, bull, and bear markets—are commonly observed. A higher value of K, such as K=4, resulted in over-segmentation, leading to reduced stability and lower silhouette scores. Thus, the decision to choose K=3 for market period classification reflects a balance between capturing key market dynamics and avoiding unnecessary complexity. The effectiveness of K=3 for financial market classification is supported by the study's findings and previous research in the field [20, 21].

4.3 Market Period Classification (K=3)

Using K=3, as determined in the previous step, the market was classified into three distinct clusters. Cluster 0 (Low-Risk Market) is characterized by a low market risk factor (Mkt-RF), high profitability (RMW), and low investment (CMA), indicating a stable and low-volatility market environment [9, 10]. Cluster 1 (Bull Market) exhibits high market returns (Mkt-RF) and a high small-cap factor (SMB), signaling a growth-driven market typically observed during bull market conditions [2, 11]. Cluster 2 (Bear/Volatile Market) is defined by low Mkt-RF, high HML (suggesting a value-driven market), and low profitability (RMW), indicating a downturn or a volatile market environment [3, 9]. This classification scheme clearly segments market regimes and offers valuable insights into the overall market behavior, capturing distinct phases such as stable, growth, and downturn market conditions [17, 19].

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4.4 Visualization of Market Clusters

To better understand the market periods and their interactions, two visualizations were created. The first is a Market Cluster 3D Visualization (see Fig. 7) using the Mkt-RF, SMB, and HML factors that clearly show the clustering based on key market characteristics. The second visualization is a 2D Market Cluster Visualization (see Figure 8) using the RMW and CMA factors that offer more detailed information about the clustering dynamics with respect to profitability and investment factors. Figure 7 and Figure 8 present the 3D and 2D market cluster visualizations, which show how each cluster is spread across the factors used in this study and where each cluster is colored differently to distinguish it.

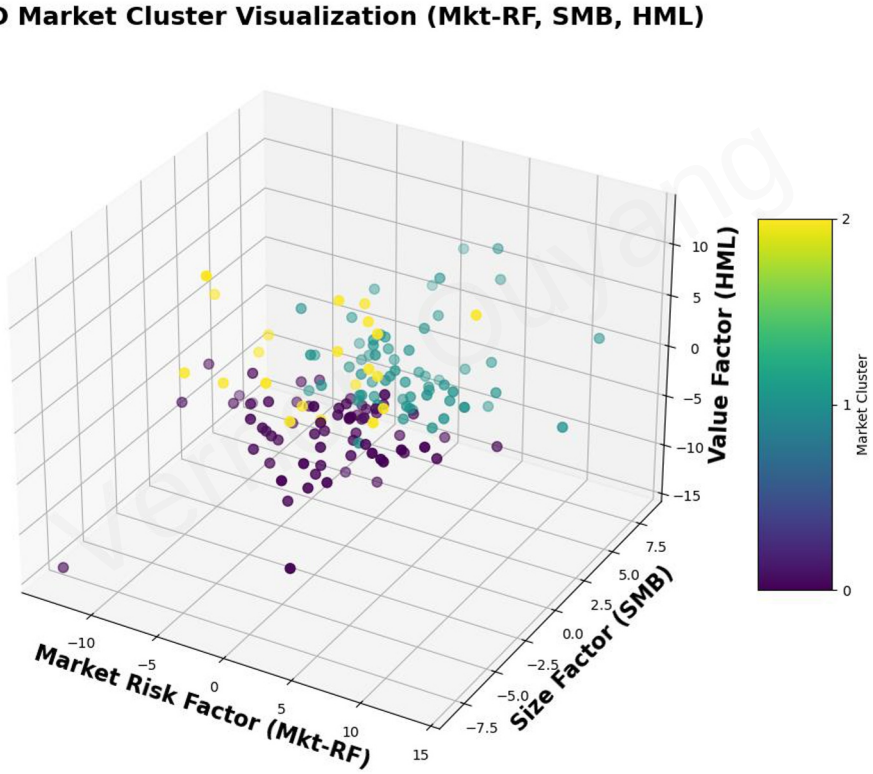

Fig. 7. 3D Market Cluster Visualization

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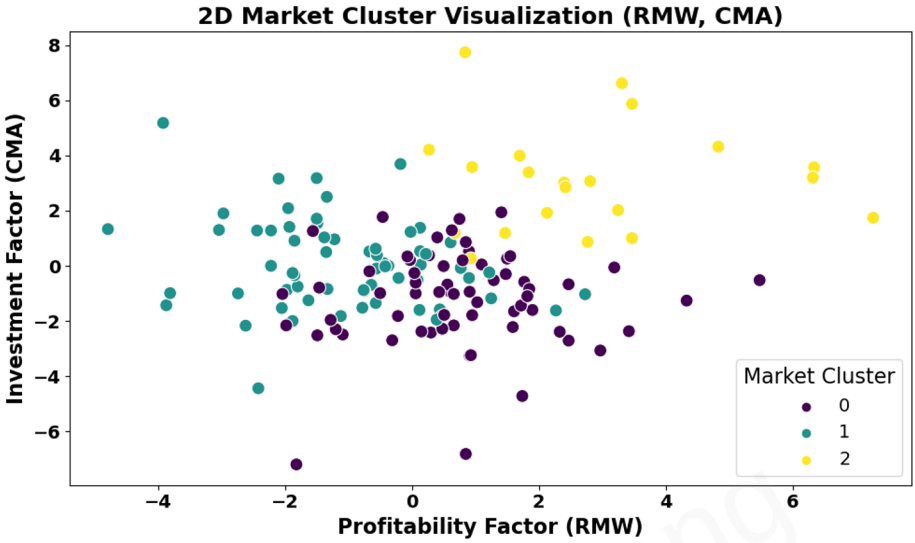

Fig. 8. 2D Market Cluster Visualization

4.5 Key Findings

The clustering analysis provided several key insights into market behavior. The K=3 clustering method effectively identified three major market regimes: Low-Risk Market, Bull Market, and Bear/Volatile Market. These regimes align with the commonly observed phases in financial markets. Moreover, the comparison between clustering techniques revealed that the Gaussian Mixture Model (GMM), which allows for soft clustering, was more effective in capturing market transitions than K-Means. This suggests that GMM is better suited for modeling the fluid and dynamic nature of market conditions. Additionally, the time-series analysis of market transitions showed distinct patterns over time, with frequent shifts between clusters, particularly during high volatility periods (see Figure 9). These results emphasize the dynamic nature of the market and confirm the appropriateness of using K=3 for market regime modeling. The identified market periods provide a valuable framework that can be applied to dynamic investment strategies, enhancing market predictions and decision-making.

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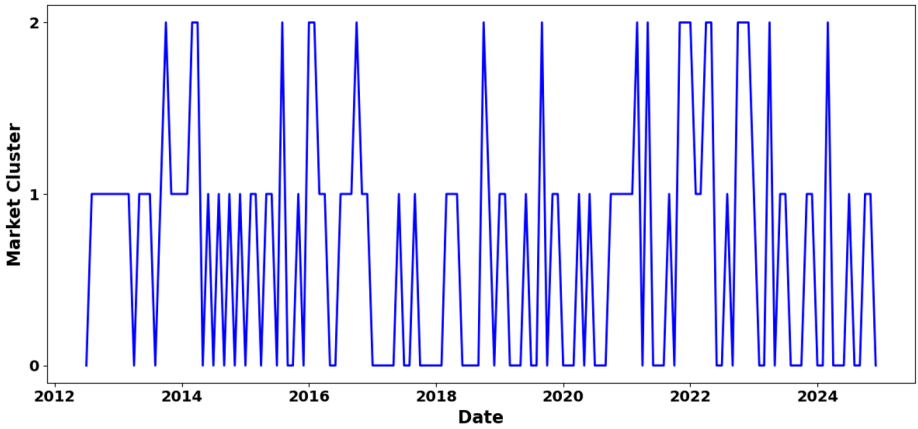

Fig. 9. Market Transitions

This study's clustering analysis corroborates existing research that advocates for the use of K=3 to segment financial market regimes. The results align with the findings of Tang and Zhang, who suggested that categorizing market periods into three segments—low-risk, bull, and bear markets—offers a clear and practical approach to understanding market behavior. Additionally, Lu's research demonstrates that K=3 provides an effective clustering solution for improving the prediction accuracy of market trends and trading strategies in the U.S. stock market.

5 Dynamic Multi-Factor Stock Selection Strategy

5.1 Data Preparation

To implement the dynamic multi-factor stock selection strategy, this study first collected relevant market data, including factors such as Mkt-RF, SMB, HML, RMW, and CMA, which were used to compute the returns of the selected stocks. The dataset was standardized to ensure all market factors were on the same scale, thus allowing for the effective comparison and combination of various factors in the stock selection process. To maintain consistency, the data from 2012 to 2024 was split into training and test sets using a 60%-40% ratio. The training set was used to fit the model, while the test set was utilized to assess the generalizability of the strategy. Additionally, market periods were assigned based on the clustering results from earlier analysis. These periods provided insights into how the market transitioned through distinct phases, which was crucial for selecting the most appropriate stocks in each phase [16-18].

5.2 Optimized Investment Portfolio (Long-Only)

The investment portfolio was optimized using a long-only constraint to ensure that only long positions were taken in the selected stocks. To balance risk and return effectively,

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the portfolio weights for the stocks were optimized using the minimum variance optimization method. This method aims to minimize the risk of the portfolio while ensuring the highest possible return within the given constraints. The stock selection was based on the market conditions identified through clustering analysis, ensuring that stocks selected for the portfolio were well-aligned with the market phase (i.e., low-risk, bull, or bear market). The results of the portfolio optimization are outlined in Table 2, which shows the stock allocations for each market cluster, providing a clear view of how different market phases affect the portfolio structure [19, 20].

Table 2. Stock Type Allocations for Each Market Cluster

Market Cluster	Growth Stocks	Neutral Stocks	Value Stocks
Low-Risk Market	48.7%	1.69%	49.98%
Bull Market	34.35%	31.73%	33.93%
Bear/Volatile Market	3.38%	69.2%	27.42%

5.3 Optimization of Stock Selection Strategy

In the dynamic stock selection strategy, stocks were chosen based on their exposure to various market factors. Each market condition was used to compute average returns, and top-performing stocks were selected. Portfolio weights were optimized using the minimum variance method to minimize risk, and performance was evaluated against the S&P 500 index [1].

5.4 Final Investment Strategies

The final investment strategies were developed using a combination of machine learning and traditional optimization techniques to achieve a robust and adaptive approach to stock selection. The first strategy, labeled the ML-Optimized Strategy (Dynamic Factor-Based Selection), dynamically adjusts the stock selection process based on the prevailing market conditions. This strategy integrates machine learning models that continuously learn from the data and adapt to changing market dynamics, making it highly flexible and responsive [4-6]. The second strategy, Minimum Variance Portfolio (Risk-Optimized Strategy), focuses on minimizing risk rather than maximizing returns. It uses historical data to determine the optimal portfolio weights for each stock that balances the trade-off between risk and return. Finally, the S&P 500 Index Benchmark serves as a passive investment strategy, providing a baseline for comparison against the more active, optimized strategies [9, 12] (See Figure 10).

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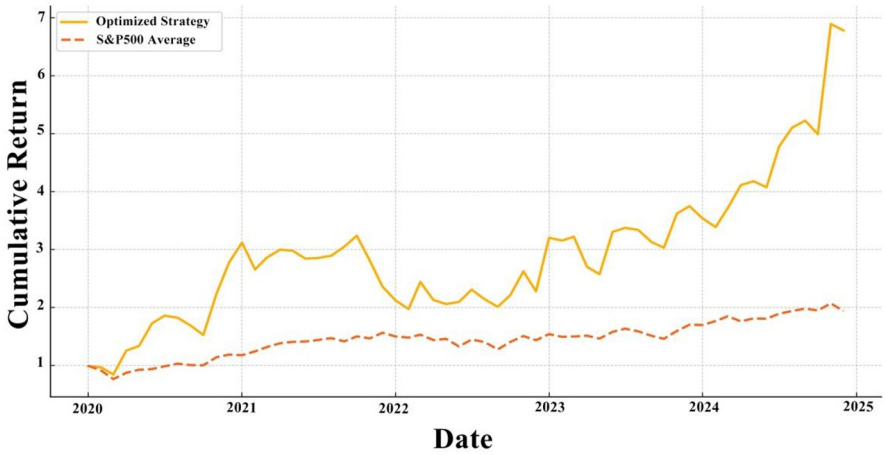

Fig. 10. Dynamic Multi-Factor Stock Selection Strategy vs S&P 500 Performance (Test Period)

5.5 Performance Comparison

To evaluate the effectiveness of the investment strategies, several performance metrics were employed, including the annualized return (CAGR), maximum drawdown, and Sharpe ratio. These metrics offer a comprehensive view of the risk-adjusted returns of the strategies. The results, summarized in Table 3, highlight that the Dynamic Strategy, which uses machine learning for dynamic stock selection, outperformed the Non-Dynamic Strategy and the S&P 500 Index in terms of CAGR, delivering an annualized return of 47.57%. In contrast, the Non-Dynamic Strategy produced a CAGR of 20.27%, while the S&P 500 Index had a return of 14.41%. The Dynamic Strategy also demonstrated better risk management, as evidenced by a lower maximum drawdown of 39.10% compared to 29.12% for the Non-Dynamic Strategy and 23.25% for the S&P 500 Index. Additionally, the Dynamic Strategy achieved a Sharpe ratio of 0.95, indicating superior risk-adjusted returns [2, 6, 8].

Table 3. Backtesting Performance Metrics

Investment Strategy	Annualized Return (CAGR)	Max Drawdown	Sharpe Ratio
Non-Dynamic Strategy	20.27%	-29.12%	0.79
S&P 500 Index	14.41%	-23.25%	0.68
Dynamic Strategy	47.57%	-39.10%	0.95

5.6 Visualizations

To further illustrate the performance of the optimized strategies, several visualizations were created comparing the Dynamic Strategy against the S&P 500 Index. Figure 11 and Figure 12 showcase the performance of the optimized portfolio over time, emphasizing the superior performance of the Dynamic Strategy. These visualizations not only

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show the return profiles of the strategies but also highlight the periods when the Dynamic Strategy significantly outperformed the S&P 500 Index. The time-series plot of the market clusters over time provides insights into how the market conditions evolved and how the portfolio adjusted accordingly. Furthermore, the 3D and 2D cluster visualizations (Figure 7 and Figure 8) offer a clear depiction of the market's segmentation based on various factors, demonstrating how the market was classified into distinct periods of low-risk, bull, and bear/volatile markets [17, 19].

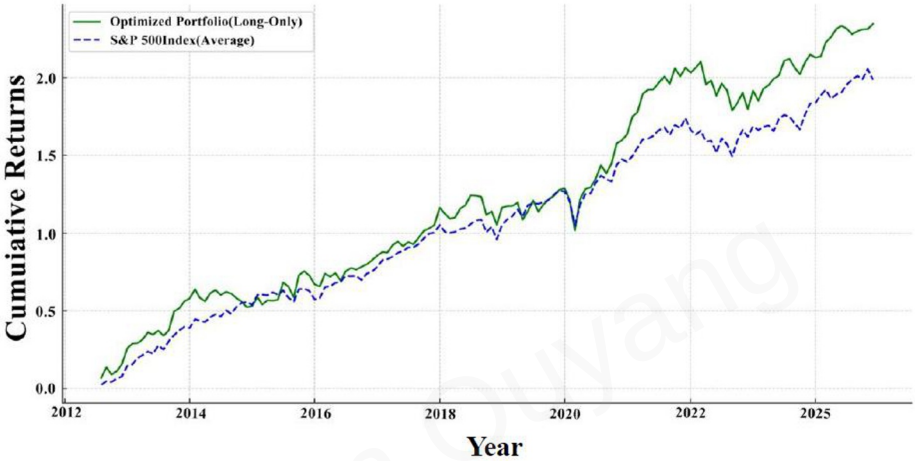

Fig. 11. Comparison of Optimized Portfolio vs S&P 500 Index

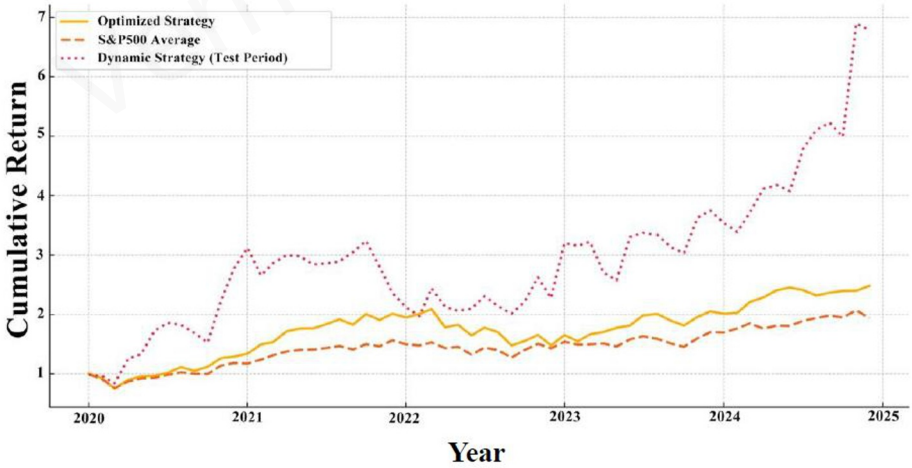

Fig. 12. Performance Comparison

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6 Performance Comparison

This section evaluates the effectiveness of different investment strategies using key financial metrics. The dynamic multi-factor stock selection strategy is compared with a non-dynamic portfolio and the S&P 500 index. The analysis includes the annualized return, maximum drawdown, and the Sharpe ratio.

6.1 Portfolio Performance Comparison

The optimized portfolio is compared to the S&P 500 Index in this paper. Figure 10 above illustrates the cumulative returns of the optimized portfolio and the S&P 500 Index, which are quite different. The optimized portfolio, which is quite volatile, outperforms the S&P 500 Index significantly, especially at the end of the test period. This means that the dynamic multi-factor stock selection strategy can identify the market trends and manage the portfolio's exposure to the key factors. Although more volatile than the S&P 500 Index, the optimized portfolio outperforms it in terms of returns. Thus, the active strategy appears more effective in the market favoring growth stocks.

6.2 Strategy Performance Comparison

More comparisons are made between the dynamic strategy, the non-dynamic strategy, and the S&P 500 Index. Fig. shows the cumulative returns of the strategies, which shows the strategies' returns during the test period from 2020 to 2024. The dynamic strategy outperforms the non-dynamic strategy and the S&P 500 Index during this period. This is because it is possible to make stock selections dynamically according to the current market conditions. The non-dynamic strategy has lower returns than the dynamic strategy, and this is especially the case when the market is more volatile. This supports the idea that dynamic factor-based selection is more efficient in changing market environments.

6.3 S&P 500 vs Optimized Strategy

The performance of the optimized strategy is compared to the S&P 500 Index in Figure 12, which shows the cumulative returns over the test period. The optimized strategy consistently outperforms the S&P 500 Index, underscoring the strength of the selected factors and their relevance in predicting stock returns under dynamic market conditions. The results also suggest that while the S&P 500 Index offers a stable return profile, the optimized strategy provides superior returns, particularly in years with heightened market volatility. This further validates the effectiveness of the dynamic multi-factor strategy in capturing market opportunities that the S&P 500 Index, as a passive investment, is unable to exploit.

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6.4 Backtesting Performance Metrics

Key performance metrics from the backtesting, such as the annualized return (CAGR), maximum drawdown, and the Sharpe ratio, are presented in Table 4. The dynamic strategy shows the highest annualized return of 47.57%, significantly outperforming both the non-dynamic strategy (20.27%) and the S&P 500 Index (14.41%). However, the dynamic strategy also experiences the highest drawdown at -39.10%, which indicates greater volatility and risk associated with this approach. The non-dynamic strategy, in contrast, offers more stable returns but with lower overall performance. The Sharpe ratio, which measures the risk-adjusted return, also supports the dynamic strategy, which achieves a value of 0.95, higher than both the non-dynamic strategy (0.79) and the S&P 500 Index (0.68). This suggests that, despite the higher risk, the dynamic strategy provides superior returns relative to the risk taken.

Table 4. Back test Performance Metrics

Strategy	Annualized Return (CAGR)	Max Drawdown	Sharpe Ratio
Non-Dynamic Strategy	20.27%	-29.12%	0.79
S&P 500 Index	14.41%	-23.25%	0.68
Dynamic Strategy	47.57%	-39.10%	0.95

6.5 Key Findings

The analysis also reveals several important insights regarding the effectiveness of the dynamic multi-factor stock selection strategy. First, the dynamic strategy outperforms both the non-dynamic strategy and the S&P 500 Index in annualized return every time. Second, the non-dynamic strategy has a more stable growth profile but has lower overall returns than the dynamic strategy. Third, the S&P 500 Index has a stable return with lower risk, but it performs worse than the dynamic strategy, which suggests that active stock selection is worthwhile in dynamic market conditions. Fourth, the risk-return tradeoff is apparent in the dynamic strategy, which has high returns but also has significant drawdowns. This is because the model was developed to select the appropriate model depending on the risk tolerance and time horizon of the investor.

7 Conclusion

The dynamic multi-factor stock selection strategy presented in this study offers substantial improvements over traditional static approaches and passive benchmarks such as the S&P 500 Index. By leveraging advanced machine learning algorithms, particularly clustering techniques and dynamic factor weighting, this strategy is designed to optimize stock selection in real-time, adapting to fluctuations in market conditions such as volatility, bull and bear phases, and economic uncertainty. Backtesting results demonstrate the superior performance of the dynamic strategy, outperforming both the non-dynamic strategies and the S&P 500 Index, highlighting its ability to capture market trends more effectively and align with investors' risk tolerance and investment horizons.

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However, despite its promising results, the proposed model is not without limitations. A primary challenge is the high level of volatility and substantial drawdowns, which underscores the need for more robust risk management strategies to mitigate risks during extreme market conditions. Additionally, the model relies on the assumption of relatively stable market conditions, an assumption that may not hold true during periods of economic uncertainty or market turmoil. The model's sensitivity to the selection of financial factors and the training period data also raises concerns about its generalizability across various market environments. These issues suggest that future research should aim to address these limitations by incorporating more sophisticated risk control measures and enhancing the model's flexibility in adapting to extreme market shifts.

Future improvements could involve the integration of advanced deep learning techniques, such as Long Short-Term Memory (LSTM) networks, which could enhance market regime classification. This would improve the model's ability to capture dynamic changes in market conditions. Additionally, reinforcement learning could be employed to optimize dynamic factor selection, further enhancing the model's adaptability and robustness in managing risks. Finally, exploring alternative portfolio optimization frameworks, such as the Black-Litterman model, could provide better risk-adjusted returns, particularly in volatile and unpredictable market environments.

In conclusion, while the dynamic multi-factor stock selection strategy shows significant promise in outperforming traditional methods, its limitations in risk management and adaptability during extreme market events need to be addressed. By refining the model through the inclusion of advanced learning techniques and improved portfolio optimization methods, future research could further enhance its effectiveness in real-world applications.

References

1. Sezer, O. B., Gudelek, M. U., Ozbayoglu, A. M.: Financial time series forecasting with deep learning: A systematic literature review: 2005–2019. Applied Soft Computing 90, 106181 (2020)
2. Gu, S., Kelly, B., Xiu, D.: Empirical asset pricing via machine learning. The Review of Financial Studies 33, 2223-2273 (2020)
3. Freyberger, J., Neuhierl, A., Weber, M.: Dissecting characteristics nonparametrically. The Review of Financial Studies 33, 2326-2377 (2020)
4. Li, J. D., Zhang, R.: Dynamic weighting multi factor stock selection strategy based on XGboost machine learning algorithm. 2018 IEEE International Conference of Safety Produce Informatization, 868-872 (2018)
5. Gao, Q., Zhou, X., Zhang, K., Huang, L., Liu, S., Zhou, F.: Incorporating Interactive Facts for Stock Selection via Neural Recursive ODEs. arXiv preprint arXiv:2210.15925 (2022)
6. Wei, Z., Dai, B., Lin, D.: E2EAI: End-to-End Deep Learning Framework for Active Investing. Proceedings of the Fourth ACM International Conference on AI in Finance, 55-63 (2023)
7. Zhong, Y., Luo, L., Wang, X., Yang J.: Multi-factor Stock Selection Model Based on Machine Learning. Engineering Letters 29(1), (2020)

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8. Fu, Y., Cao, S., Pang, T.: A Sustainable Quantitative Stock Selection Strategy Based on Dynamic Factor Adjustment. Sustainability 12(10), 3978 (2020)
9. Kelly, B. T., Pruitt, S., Su, Y.: Characteristics are covariances: A unified model of risk and return. Journal of Financial Economics 34(3), 501-524 (2019)
10. Feng, G., Giglio, S., Xiu, D.: Taming the factor zoo: A test of new factors. The Journal of Finance 75(3), 1327-1370 (2020)
11. Fama, E. F., French, K. R.: A five-factor asset pricing model. Journal of Financial Economics 116(1), 1-22 (2015)
12. Gu, S., Kelly, B., Xiu, D.: Autoencoder asset pricing models. Journal of Econometrics 222(1), 429-450 (2021)
13. Chinco, A., Clark - Joseph, A. D., Ye, M.: Sparse signals in the cross - section of returns. The Journal of Finance 74(1), 449-492 (2019)
14. Feng, G., He, J., Polson, N. G., Xu, J.: Deep learning in characteristics-sorted factor models. Journal of Financial and Quantitative Analysis 59(7), 3001-3036 (2024)
15. Fischer, T., Krauss, C.: Deep learning with long short-term memory networks for financial market predictions. European journal of operational research 270(2), 654-669 (2018)
16. Imajo, K., Minami, K., Ito, K., Nakagawa, K.: Deep portfolio optimization via distributional prediction of residual factors. Proceedings of the AAAI conference on artificial intelligence 35(1), 213-222 (2021)
17. Sörensen, K., Pardalos, P. M.: Clustering in Financial Markets. Models, Algorithms and Technologies for Network Analysis, Springer International Publishing, (2016)
18. Lu, Y.: Application of Clustering Methods to Trading Strategies in the US Equity Market. (2018)
19. Rousseeuw, P. J.: Silhouettes: a graphical aid to the interpretation and validation of cluster analysis. Journal of computational and applied mathematics 20, 53-65 (1987)
20. Kaufmann, L.: Clustering by Means of medoids. Proc. Statistical Data Analysis Based on the L1 Norm Conference, Neuchatel, 405-416 (1987)
21. Bishop, C. M., Nasrabadi, N. M.: Pattern recognition and machine learning. Springer, (2006)

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